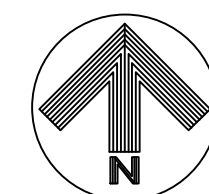
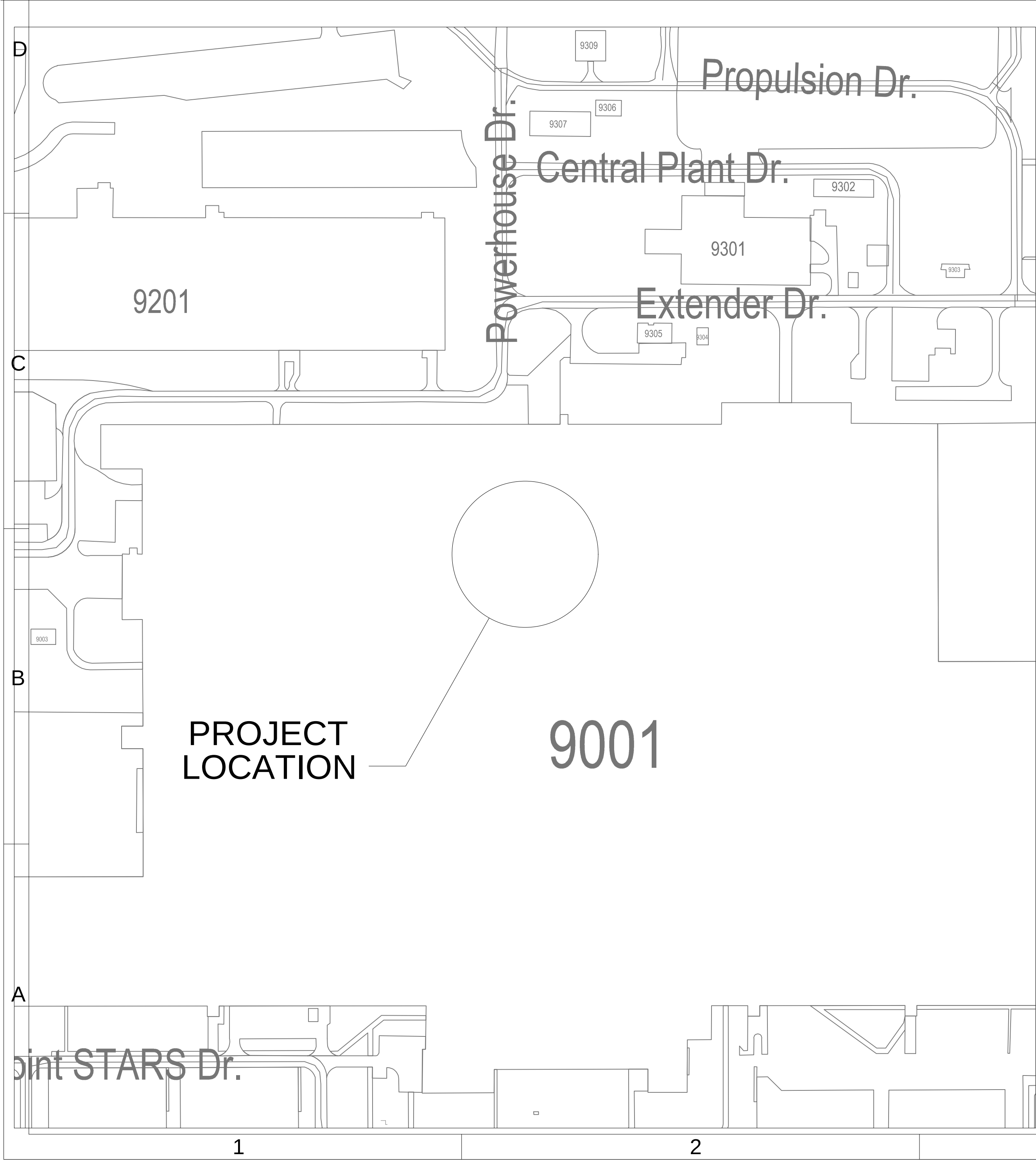
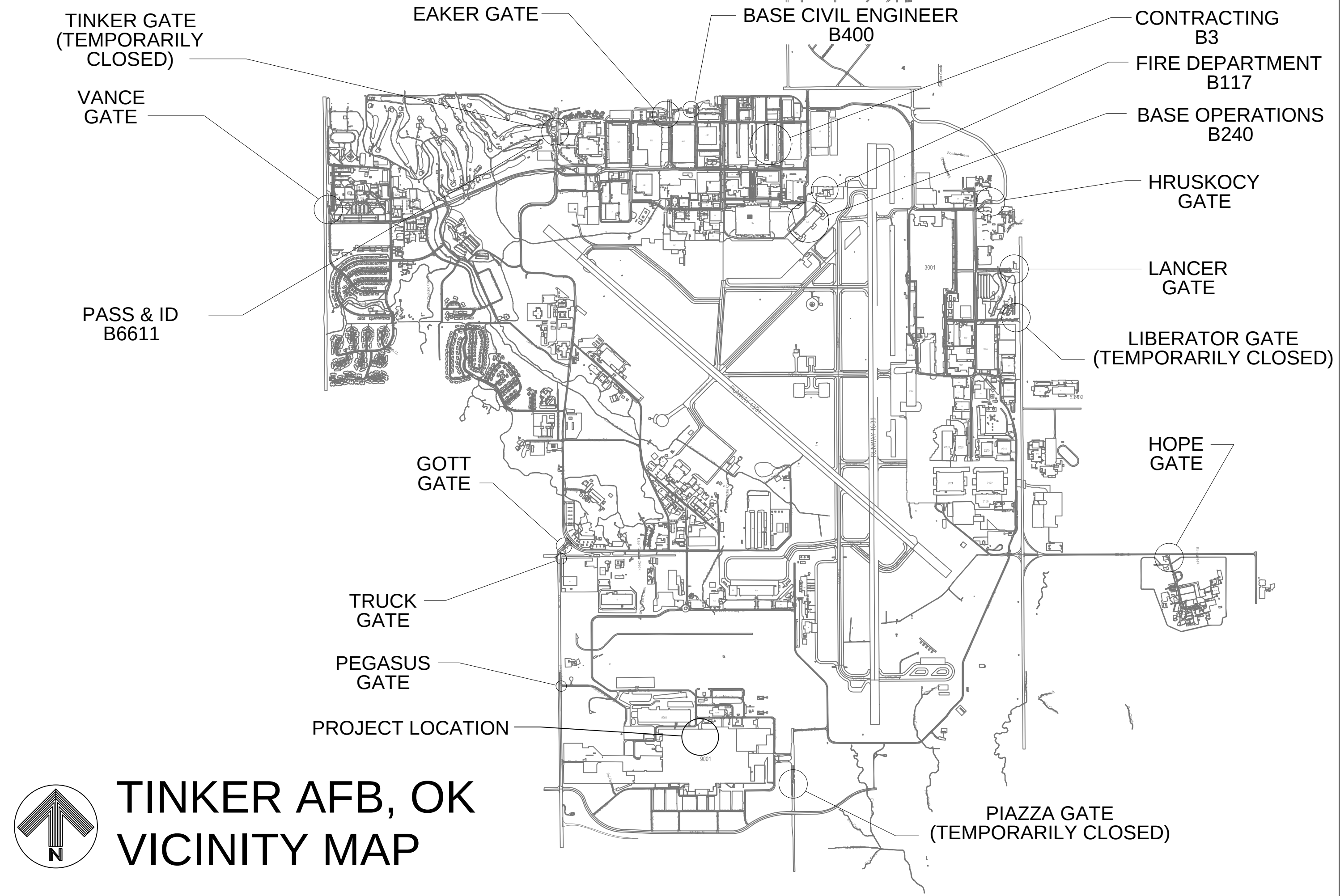


	1	2
E	WWYK150083 REPAIR TRANSFORMER VAULT 6, B9001	



**TINKER AFB, OK
VICINITY MAP**



PROJECT DRAWING INDEX	SIGNATURES
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SHT.	SHEET TITLE	SHT.	SHEET TITLE		
G-001	SHEET INDEX, VICINITY MAP AND BUILDING KEY PLAN			CORROSION ENGINEER: (CECE)	DATE:
	ARCHITECTURAL				
A-101	ARCHITECTURAL FLOOR AND ROOF PLANS, ELEVATIONS IMAGES			CE OPERATIONS: (CECO)	DATE:
	STRUCTURAL				
S-001	STRUCTURAL GENERAL NOTES			ZONE MANAGER:	DATE:
S-101	STRUCTURAL PLAN - NEW WORK				
	ELECTRICAL			CONSTRUCTION MANAGEMENT: (CECE)	DATE:
E-001	ELECTRICAL LEGEND, SYMBOLS AND ABBREVIATIONS				
ED101	ELECTRICAL PLAN - DEMOLITION			FIRE DEPARTMENT: (CEFT)	DATE:
ED601	ELECTRICAL ONE-LINE DIAGRAM - DEMOLITION				
E-101	ELECTRICAL PLAN - NEW WORK				
E-501	ELECTRICAL DETAILS			SAFETY:	DATE:
E-601	ELECTRICAL ONE-LINE DIAGRAM - NEW WORK				
E-602	ELECTRICAL SCHEDULES			INDUSTRIAL HYGIENE:	DATE:
	FIRE PROTECTION				
F-101	FIRE ALARM - DEMOLITION AND NEW WORK PLAN			ENVIRONMENTAL:	DATE:
				ASBESTOS EVALUATION:	DATE:
				COMMUNICATIONS: (COMM)	DATE:
				USING AGENCY:	DATE:
				REVIEWED BY:	DATE:
				REVIEWED BY:	DATE:
				DESIGN MANAGER: (CECE)	DATE:
				CHIEF DESIGN BRANCH: (CECE)	DATE:
				CHIEF ENGINEER DIR: (CECE)	DATE:
				BASE CIVIL ENGINEER: (CE)	DATE:

SIGNATURES

CORROSION ENGINEER: (CECE)	DATE:
CE OPERATIONS: (CECO)	DATE:
ZONE MANAGER:	DATE:
CONSTRUCTION MANAGEMENT: (CECE)	DATE:
FIRE DEPARTMENT: (CEFT)	DATE:
SAFETY:	DATE:
INDUSTRIAL HYGIENE:	DATE:
ENVIRONMENTAL:	DATE:
ASBESTOS EVALUATION:	DATE:
COMMUNICATIONS: (COMM)	DATE:
USING AGENCY:	DATE:
REVIEWED BY:	DATE:
REVIEWED BY:	DATE:
DESIGN MANAGER: (CECE)	DATE:
CHIEF DESIGN BRANCH: (CECE)	DATE:
CHIEF ENGINEER DIR: (CECE)	DATE:
BASE CIVIL ENGINEER: (CE)	DATE:



WILLIAM L. TINKER AFB, OK

WWYK150083
REPAIR TRANSFORMER
VAULT 6, B9001

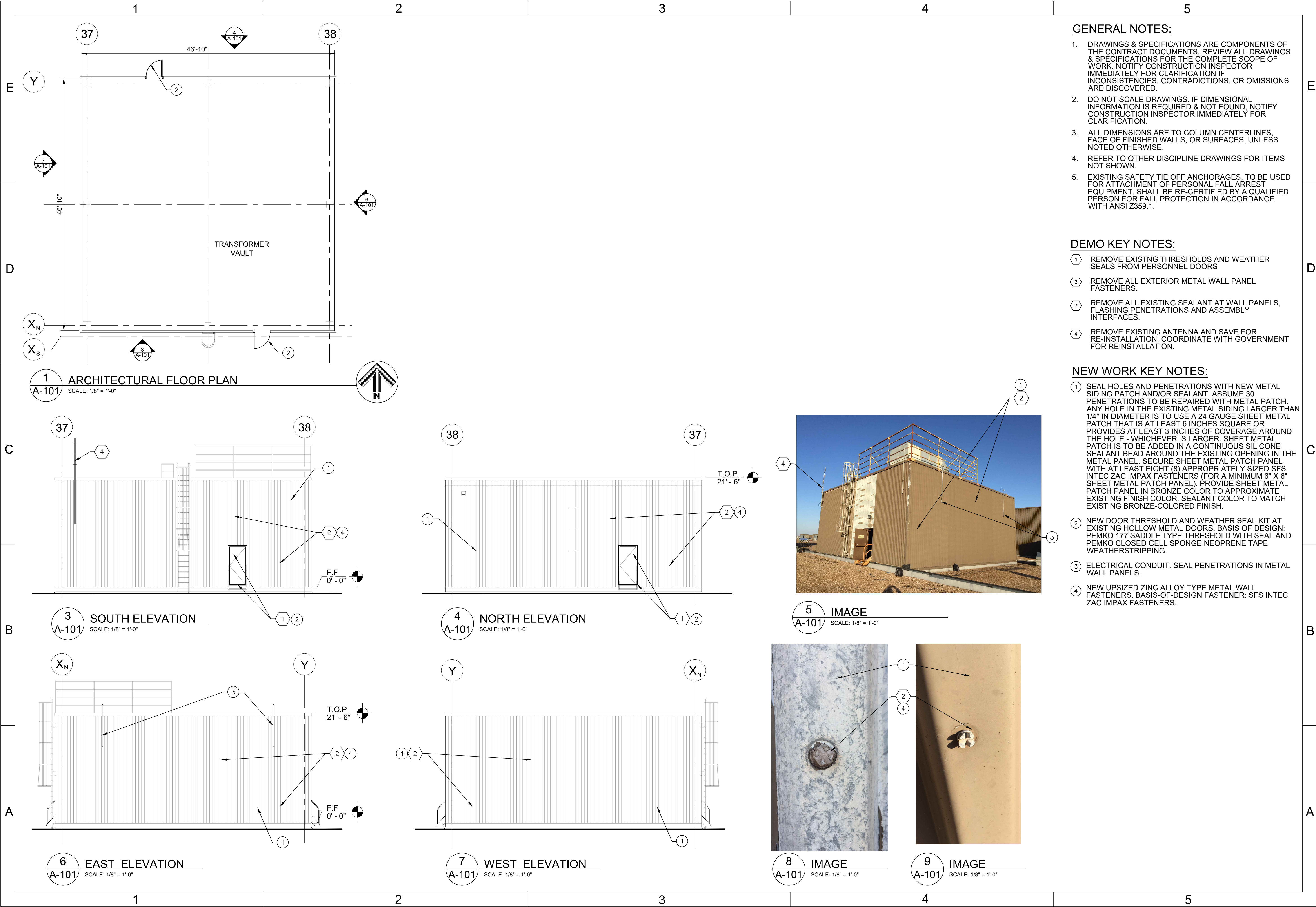
GENERAL
SHEET INDEX
VICINITY MAP AND
BUILDING KEY PLAN

SHEET NUMBER —

G-001

SHEET NUMBER _____

G-001



GENERAL NOTES:

- DRAWINGS & SPECIFICATIONS ARE COMPONENTS OF THE CONTRACT DOCUMENTS. REVIEW ALL DRAWINGS & SPECIFICATIONS FOR THE COMPLETE SCOPE OF WORK. NOTIFY CONSTRUCTION INSPECTOR IMMEDIATELY FOR CLARIFICATION IF INCONSISTENCIES, CONTRADICTIONS, OR OMISSIONS ARE DISCOVERED.
- DO NOT SCALE DRAWINGS. IF DIMENSIONAL INFORMATION IS REQUIRED & NOT FOUND, NOTIFY CONSTRUCTION INSPECTOR IMMEDIATELY FOR CLARIFICATION.
- ALL DIMENSIONS ARE TO COLUMN CENTERLINES, FACE OF FINISHED WALLS, OR SURFACES, UNLESS NOTED OTHERWISE.
- REFER TO OTHER DISCIPLINE DRAWINGS FOR ITEMS NOT SHOWN.
- EXISTING SAFETY TIE OFF ANCHORAGES, TO BE USED FOR ATTACHMENT OF PERSONAL FALL ARREST EQUIPMENT, SHALL BE RE-CERTIFIED BY A QUALIFIED PERSON FOR FALL PROTECTION IN ACCORDANCE WITH ANSI Z359.1.

DEMO KEY NOTES:

- REMOVE EXISTING THRESHOLDS AND WEATHER SEALS FROM PERSONNEL DOORS
- REMOVE ALL EXTERIOR METAL WALL PANEL FASTENERS.
- REMOVE ALL EXISTING SEALANT AT WALL PANELS, FLASHING PENETRATIONS AND ASSEMBLY INTERFACES.
- REMOVE EXISTING ANTENNA AND SAVE FOR RE-INSTALLATION. COORDINATE WITH GOVERNMENT FOR REINSTALLATION.

NEW WORK KEY NOTES:

- SEAL HOLES AND PENETRATIONS WITH NEW METAL SIDING PATCH AND/OR SEALANT. ASSUME 30 PENETRATIONS TO BE REPAIRED WITH METAL PATCH. ANY HOLE IN THE EXISTING METAL SIDING LARGER THAN 1/4" IN DIAMETER IS TO USE A 24 GAUGE SHEET METAL PATCH THAT IS AT LEAST 6 INCHES SQUARE OR PROVIDES AT LEAST 3 INCHES OF COVERAGE AROUND THE HOLE - WHICHEVER IS LARGER. SHEET METAL PATCH IS TO BE ADDED IN A CONTINUOUS SILICONE SEALANT BEAD AROUND THE EXISTING OPENING IN THE METAL PANEL. SECURE SHEET METAL PATCH PANEL WITH AT LEAST EIGHT (8) APPROPRIATELY SIZED SFS INTEC ZAC IMPAX FASTENERS (FOR A MINIMUM 6" X 6" SHEET METAL PATCH PANEL). PROVIDE SHEET METAL PATCH PANEL IN BRONZE COLOR TO APPROXIMATE EXISTING FINISH COLOR. SEALANT COLOR TO MATCH EXISTING BRONZE-COLORED FINISH.
- NEW DOOR THRESHOLD AND WEATHER SEAL KIT AT EXISTING HOLLOW METAL DOORS. BASIS OF DESIGN: PEMKO 177 SADDLE TYPE THRESHOLD WITH SEAL AND PEMKO CLOSED CELL SPONGE NEOPRENE TAPE WEATHERSTRIPPING.
- ELECTRICAL CONDUIT. SEAL PENETRATIONS IN METAL WALL PANELS.
- NEW UPSIZED ZINC ALLOY TYPE METAL WALL FASTENERS, BASIS-OF-DESIGN FASTENER: SFS INTEC ZAC IMPAX FASTENERS.

AIR FORCE MATERIEL COMMAND
72 ABW CIVIL ENGINEER DIRECTORATE
TINKER AFB, OK

PROJECT INFORMATION

WWYK150083
REPAIR TRANSFORMER
VAULT 6, B9001
TINKER AIR FORCE BASE, OKLAHOMA

REVISIONS	MARK	DATE	DESCRIPTION
1			
2			
3			
4			

MANAGEMENT INFORMATION

ISSUE DATE: 12 DEC 2025
DESIGN GROUP: HDR
DESIGNED BY: AM
DRAWN BY: JS
CHECKED BY: RA
DESIGN FILE LOCATION:

SHEET TITLE

ARCHITECTURAL
FLOOR AND ROOF
PLANS, ELEVATIONS
IMAGES

SHEET NUMBER

A-101

1

2

3

4

5

GENERAL STRUCTURAL NOTES (GSN)

GENERAL

G1. SCOPE

THE NOTES ON THIS SHEET AND THE STANDARD STRUCTURAL DETAILS ARE GENERAL AND APPLY TO THE ENTIRE PROJECT WHETHER SPECIFICALLY CALLED OUT OR NOT, EXCEPT WHERE THERE ARE SPECIFIC INDICATIONS TO THE CONTRARY ON STRUCTURAL SHEETS. IF THERE ARE QUESTIONS, THEY SHALL BE SUBMITTED TO THE STRUCTURAL ENGINEER AND ANSWERED IN WRITING PRIOR TO CONSTRUCTION.

G2. APPLICABLE SPECIFICATIONS AND CODES

A. UFC 1-200-01 WITH CHANGE 4, DATED 17 DECEMBER 2024
B. UFC 3-301-01 WITH CHANGE 4, DATED 3 JUNE 2025
C. INTERNATIONAL BUILDING CODE, IBC 2024 WITH APPLICABLE EDITIONS OF THE CODE REFERENCED STANDARDS.
D. TINKER AIR FORCE BASE FACILITY STANDARDS
E. ACI 318-19
F. ASCE 7-22
G. INTERNATIONAL EXISTING BUILDING CODE, IEBG 2024

G3. EXISTING DRAWINGS

THE CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL. THE PLANS AND DETAILS FOR THE EXISTING STRUCTURES PRESENTED IN THE DRAWINGS ARE FOR REFERENCE ONLY BASED ON THE FOLLOWING AS-BUILT DRAWINGS:
1. NEW MANUFACTURING PLANT FOR OKLAHOMA INDUSTRIES AUTHORITY TENANCY BY GM ASSEMBLY DIVISION DATED NOVEMBER 1977.

G4. DESIGN CRITERIA

1. APPLIES TO ALL STRUCTURES (UNO)
A. DEAD LOAD:
1. ACTUAL TRIBUTARY STRUCTURE WEIGHT
2. ELECTRICAL EQUIPMENT SELF-WEIGHT
B. LIVE LOAD:
1. VAULT: 100 PSF
2. ROOF: 20 PSF (NON-REDUCIBLE)
C. WIND:
1. RISK CATEGORY: III
2. ULTIMATE WIND SPEED; V_{ULT} : 115 MPH
3. ALLOWABLE STRESS WIND SPEED; V_{ASD} : 89 MPH
4. EXPOSURE: C
5. DESIGN WIND PRESSURE: 28 PSF
6. TORNADO WIND SPEED (V_T) = 80 MPH
7. TORNADO WIND PRESSURE = 16 PSF
D. SEISMIC:
1. RISK CATEGORY: III
2. IMPORTANCE FACTOR: 1.25
3. SPECTRAL RESPONSE ACCELERATION, S_S = 0.32g
4. SPECTRAL RESPONSE ACCELERATION, S_1 = 0.08g
5. SITE CLASS: D
6. SEISMIC DESIGN CATEGORY: B
7. SPECTRAL RESPONSE ACCELERATION, S_{DS} = 0.23g
8. SPECTRAL RESPONSE ACCELERATION, S_{D1} = 0.11g
9. BASIC SEISMIC FORCE RESISTING SYSTEM: SYSTEMS NOT SPECIFICALLY DETAILED FOR SEISMIC RESISTANCE ASCE CHAPTER 13 - NON-STRUCTURAL COMPONENTS
10. ANALYSIS PROCEDURE:
11. SEISMIC DESIGN FORCE (MAX): $1.6 \cdot S_{DS} \cdot I_p \cdot \text{WEIGHT}$
12. BUILDING RESPONSE MODIFICATION FACTOR, R_p = 3.5
13. COMPONENT IMPORTANCE FACTOR, I_p = 1.0
14. STRUCTURE DUCTILITY REDUCTION FACTOR (R_u): 1.3
15. COMPONENT RESONANCE DUCTILITY FACTOR (C_{dR}): 1.0
16. COMPONENT STRENGTH FACTOR (R_{po}): 1.5
E. SNOW LOAD:
1. GROUND SNOW LOAD; P/g : 20 PSF
2. FROST DEPTH: 18 INCHES
F. ICE LOADS:
1. ICE THICKNESS: 1.74 INCHES
2. ICE THICKNESS HEIGHT FACTOR: 1.002
3. DESIGN ICE THICKNESS: 1.74 INCHES
4. DESIGN WEIGHT OF ICE: 8.14 PSF
G. ROOF RAIN LOAD:
1. 15-MINUTE PRECIPITATION INTENSITY: 8.66 IN/HR
2. 60-MINUTE PRECIPITATION INTENSITY: 4.41 IN/HR

G5. SAFETY

SAFETY AND STRUCTURE STABILITY DURING CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. STRUCTURES HAVE BEEN DESIGNED TO RESIST THE DESIGN LIVE LOADS ONLY AS A COMPLETED STRUCTURE.

G6. OPENINGS

OPENINGS FOR PIPES, DUCTS, CONDUITS, ETC. ARE NOT ALL SHOWN ON THE STRUCTURAL DRAWINGS. COORDINATE AND PROVIDE OPENINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

G7. SPECIAL INSPECTIONS

THE CONTRACTOR MUST EMPLOY ONE OR MORE APPROVED AGENCIES TO PERFORM INSPECTIONS DURING CONSTRUCTION ON THE TYPES OF WORK LISTED UNDER SECTION 1705 OF THE IBC REQUIRING VERIFICATION AND INSPECTION. THESE INSPECTIONS ARE IN ADDITION TO THE INSPECTIONS DEFINED IN SECTION 110. THE INSPECTING AGENCY MUST PROVIDE REPORTS OF THE SPECIAL INSPECTIONS DIRECTLY TO THE CONTRACTING OFFICER.

G8. STANDARD DETAILS

THE STANDARD DETAILS DEPICT TYPICAL DETAILING TO BE USED ON THIS PROJECT. IF CONDITIONS ARE NOT EXPLICITLY SHOWN ON THE DRAWINGS THEY SHALL BE MADE SIMILAR TO THE STANDARD DETAILS. OBTAIN APPROVAL OF ENGINEER IN WRITING FOR SIMILAR CONDITIONS PRIOR TO CONSTRUCTION.

G9.

THE CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL.

G10.

CONTRACTOR TO SUBMIT FOR REVIEW ALL EQUIPMENT SIZES, OPERATING WEIGHTS, VIBRATION FORCES, SUPPORT LOCATIONS, ALONG WITH ANY FLOOR OPENINGS, NOTCHES, AND RECESSES REQUIRED BY SUCH EQUIPMENT. CONCRETE SUPPORT PADS AND/OR FRAMING REQUIRED TO SUPPORT SAID EQUIPMENT SHALL NOT BE FABRICATED AND PLACED UNTIL THE CONCRETE SUPPORT PADS AND/OR FRAMING IS APPROVED TO SUPPORT THE EQUIPMENT.

POST INSTALLED ANCHORS

P1.

POST-INSTALLED ANCHORS SHALL INCLUDE ADHESIVE ANCHORS (THREADED RODS, BOLTS OR REINFORCING BARS), EXPANSION ANCHORS, AND UNDERCUT ANCHORS INSTALLED INTO HARDENED CONCRETE.

P2.

POST-INSTALLED ANCHORS SHALL ONLY BE USED WHERE INDICATED ON THE DRAWINGS. CONTRACTOR SHALL OBTAIN APPROVAL FROM ENGINEER PRIOR TO USING POST-INSTALLED ANCHORS FROM MISSING OR MISPLACED CAST-IN-PLACE ANCHORS.

P3.

CARE SHOULD BE TAKEN TO AVOID CONFLICTS WITH EXISTING REINFORCING STEEL AND OTHER EMBEDDED ITEMS WHEN DRILLING HOLES. REINFORCING BARS SHALL NOT BE DAMAGED DURING DRILLING OR ANCHOR INSTALLATION. HOLES SHALL BE DRILLED AND CLEANED PER THE PRODUCT MANUFACTURER'S INSTRUCTIONS. ANCHORS SHALL BE INSTALLED PER THE PRODUCT MANUFACTURER'S INSTRUCTIONS AT NOT LESS THAN MINIMUM EDGE DISTANCES AND/OR SPACING INDICATED IN THE MANUFACTURER'S LITERATURE.

P4.

SUBSTITUTION REQUESTS FOR PRODUCTS OTHER THAN THOSE INDICATED ON THE DRAWINGS SHALL BE SUBMITTED TO ENGINEER FOR REVIEW AND APPROVAL. CURRENT PRODUCT ICC-ESR EVALUATION REPORTS SHALL BE INCLUDED WITH THE SUBMITTAL PACKAGE. IF METHODS AND PROCEDURES REQUIRED BY THE BUILDING CODE MAY BE REQUIRED AS PART OF THE SUBMITTAL PACKAGE.

P5.

UNLESS NOTES OTHERWISE, THE MINIMUM EMBEDMENT PROVIDED FOR ADHESIVE ANCHORED REINFORCING BARS SHALL DEVELOP THE FULL TENSILE STRENGTH OF THE BARS.

P6.

SPECIAL INSPECTION WILL BE PROVIDED FOR ALL POST-INSTALLED ANCHORS AS INDICATED IN SPECIAL INSPECTION REQUIREMENTS PER SPECIFICATION SECTION 05 05 20.

STEEL

S1.

DESIGN STRENGTHS:
ALL OTHER PLATES AND SHAPES: $F_y=36$ KSI

S2.

DIMENSIONS:
TO CENTERLINES OF COLUMNS AND BEAMS, TOP SURFACES OF BEAMS AND TUBES AND BACKS OF CHANNELS AND ANGLES UNO.

S3.

ELEVATIONS:
TOP OF STEEL REFERS TO TOP SURFACE OF MEMBER OR FLANGE UNO.

S4.

WHEN FILLET WELD SIZE IS NOT INDICATED, PROVIDE MAXIMUM WELD SIZE BASED ON MATERIAL THICKNESS IN ACCORDANCE WITH AISC SPECIFICATIONS.

S5.

ALL BOLTED STRUCTURAL CONNECTIONS ARE BEARING TYPE CONNECTIONS UNLESS OTHERWISE SPECIFIED TO BE SLIP-CRITICAL. PROVIDE LOAD INDICATING WASHERS AT SLIP-CRITICAL CONNECTIONS.

S6.

CONFORM TO AISC 360, STEEL CONSTRUCTION MANUAL AND AISC 341, SEISMIC DESIGN MANUAL.

MODIFICATIONS TO EXISTING CONCRETE:

EC1.

DRILLED HOLE FOR DOWEL SHALL BE THOROUGHLY CLEANED BY WIRE BRUSHING AND AIR PRESSURE OR VACUUM TO REMOVE ALL DRILLING DUST. THE HOLE SHALL THEN BE PARTIALLY FILLED WITH EPOXY AND THE UNCOATED BAR INSERTED, TWISTED AND LEFT IN PLACE.

EC2.

AT LOCATIONS WHERE NEW PENETRATIONS FOR CONDUITS ARE REQUIRED CONTRACTOR SHALL NON-DESTRUCTIVELY LOCATED EXISTING REINFORCING PRIOR TO DRILLING HOLES. LOCATE CONDUITS TO NOT DAMAGE REINFORCING.

EC3.

NEW OPENINGS IN EXISTING CONCRETE SHALL BE CUT 1 INCH OVERSIZE, COATED WITH EPOXY BONDING COMPOUND, AND MORTAR FINISHED TO THE REQUIRED FINISH SIZE.

EC4.

STRUCTURAL DIMENSIONS RELATED TO OR CONTROLLED BY EXISTING STRUCTURES SHALL BE VERIFIED IN FIELD BY THE CONTRACTOR PRIOR TO CONCRETE WORK.

DEFERRED SUBMITTALS:

1. DEFINITION:

DELEGATE DESIGN SUBMITTALS ARE ITEMS FOR WHICH THE FINAL DESIGN OF A SPECIFIC ELEMENT OF THE PERMANENT CONSTRUCTION WORK EXPRESSLY DELEGATED TO A PROPERLY LICENSED AND REGISTERED DESIGN PROFESSIONAL LICENSED IN THE STATE OF OKLAHOMA RETAINED OR EMPLOYED BY THE CONTRACTOR, SUBCONTRACTOR, SUPPLIER OF ANY TIER. "DELEGATED DESIGN" APPLIES ONLY TO PERMANENT WORK THAT WILL BE PART OF THE COMPLETED PROJECT AS A FUNCTIONING WHOLE.

2. DELEGATED DESIGN LIST:

THE FOLLOWING IS A LIST OF DELEGATED SUBMITTALS THAT ARE EXPECTED TO CONTAIN STAMPED STRUCTURAL CALCULATIONS OF SAFETY-RELATED SYSTEM INFORMATION FOR REVIEW. THESE SUBMITTALS APPLY TO DELEGATED DESIGN SYSTEMS PRIOR TO INSTALLATION OF THE INDICATED STRUCTURAL ELEMENT, EQUIPMENT, DISTRIBUTION SYSTEM, COMPONENT OR ITS ANCHORAGE. THE CONTRACTOR SHALL SUBMIT THE REQUIRED ENGINEER CERTIFICATION, SUPPORTING DATA, AND DRAWINGS FOR REVIEW AND ACCEPTANCE BY THE CONTRACTING OFFICER.

DEFERRED SUBMITTALS

SPEC SECTIONS

ITEM

VARIOUS

ANY EQUIPMENT OR COMPONENT IN WHICH A TECHNICAL SPECIFICATION REQUIRES SUBMITTAL OF EQUIPMENT OR ANCHORAGE SYSTEM CALCULATIONS

E

D

C

B

A

E

D

C

B

A

PROJECT INFORMATION

WWYK150083
REPAIR TRANSFORMER
VAULT 6, B9001
TINKER AIR FORCE BASE, OKLAHOMA

REVISIONS

MARK

DATE

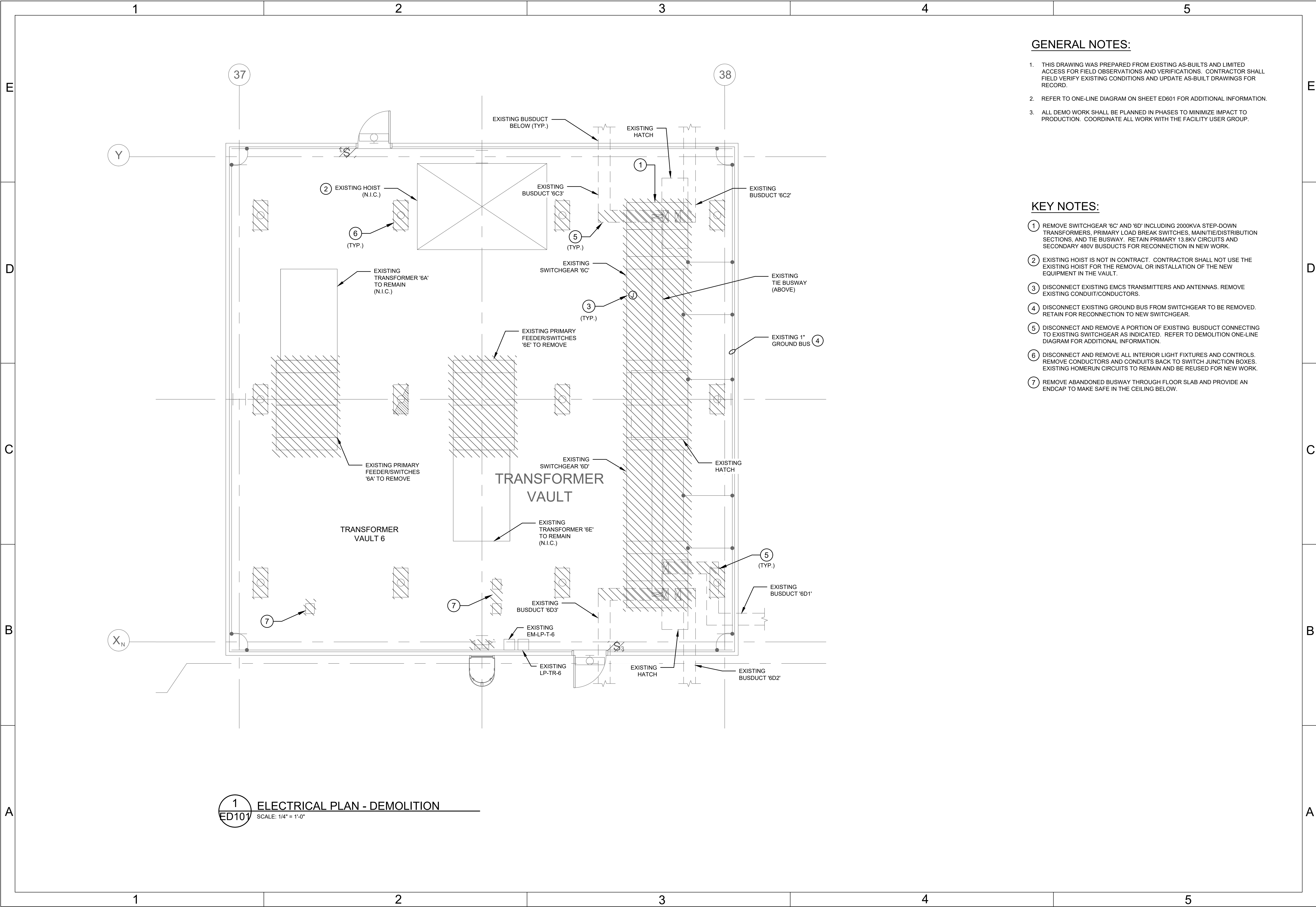
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ISSUE DATE: 12 DEC 2025
DESIGN GROUP: HDR
DESIGNED BY: CM
DRAWN BY: RN
CHECKED BY: KT
DESIGN FILE LOCATION:

MANAGEMENT INFORMATION

STRUCTURAL
GENERAL NOTES

SHEET NUMBER:
S-001



GENERAL NOTES:

1. THIS DRAWING WAS PREPARED FROM EXISTING AS-BUILTS AND LIMITED ACCESS FOR FIELD OBSERVATIONS AND VERIFICATIONS. CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS AND UPDATE AS-BUILT DRAWINGS FOR RECORD.
2. REFER TO ONE-LINE DIAGRAM ON SHEET ED601 FOR ADDITIONAL INFORMATION.
3. ALL DEMO WORK SHALL BE PLANNED IN PHASES TO MINIMIZE IMPACT TO PRODUCTION. COORDINATE ALL WORK WITH THE FACILITY USER GROUP.

KEY NOTES:

- 1 REMOVE SWITCHGEAR '6C' AND '6D' INCLUDING 2000KVA STEP-DOWN TRANSFORMERS, PRIMARY LOAD BREAK SWITCHES, MAIN/TIE/DISTRIBUTION SECTIONS, AND TIE BUSWAY. RETAIN PRIMARY 13.8KV CIRCUITS AND SECONDARY 480V BUSDUCTS FOR RECONNECTION IN NEW WORK.
- 2 EXISTING HOIST IS NOT IN CONTRACT. CONTRACTOR SHALL NOT USE THE EXISTING HOIST FOR THE REMOVAL OR INSTALLATION OF THE NEW EQUIPMENT IN THE VAULT.
- 3 DISCONNECT EXISTING EMCS TRANSMITTERS AND ANTENNAS. REMOVE EXISTING CONDUIT/CONDUCTORS.
- 4 DISCONNECT EXISTING GROUND BUS FROM SWITCHGEAR TO BE REMOVED. RETAIN FOR RECONNECTION TO NEW SWITCHGEAR.
- 5 DISCONNECT AND REMOVE A PORTION OF EXISTING BUSDUCT CONNECTING TO EXISTING SWITCHGEAR AS INDICATED. REFER TO DEMOLITION ONE-LINE DIAGRAM FOR ADDITIONAL INFORMATION.
- 6 DISCONNECT AND REMOVE ALL INTERIOR LIGHT FIXTURES AND CONTROLS. REMOVE CONDUCTORS AND CONDUITS BACK TO SWITCH JUNCTION BOXES. EXISTING HOMERUN CIRCUITS TO REMAIN AND BE REUSED FOR NEW WORK.
- 7 REMOVE ABANDONED BUSWAY THROUGH FLOOR SLAB AND PROVIDE AN ENDCAP TO MAKE SAFE IN THE CEILING BELOW.



AIR FORCE MATERIEL COMMAND
72 ABW CIVIL ENGINEER DIRECTORATE

TINKER AFB, OK



PROJECT INFORMATION

WWYK150083
REPAIR TRANSFORMER
VAULT 6, B9001

TINKER AIR FORCE BASE, OKLAHOMA

REVISIONS	MARK	DATE	DESCRIPTION

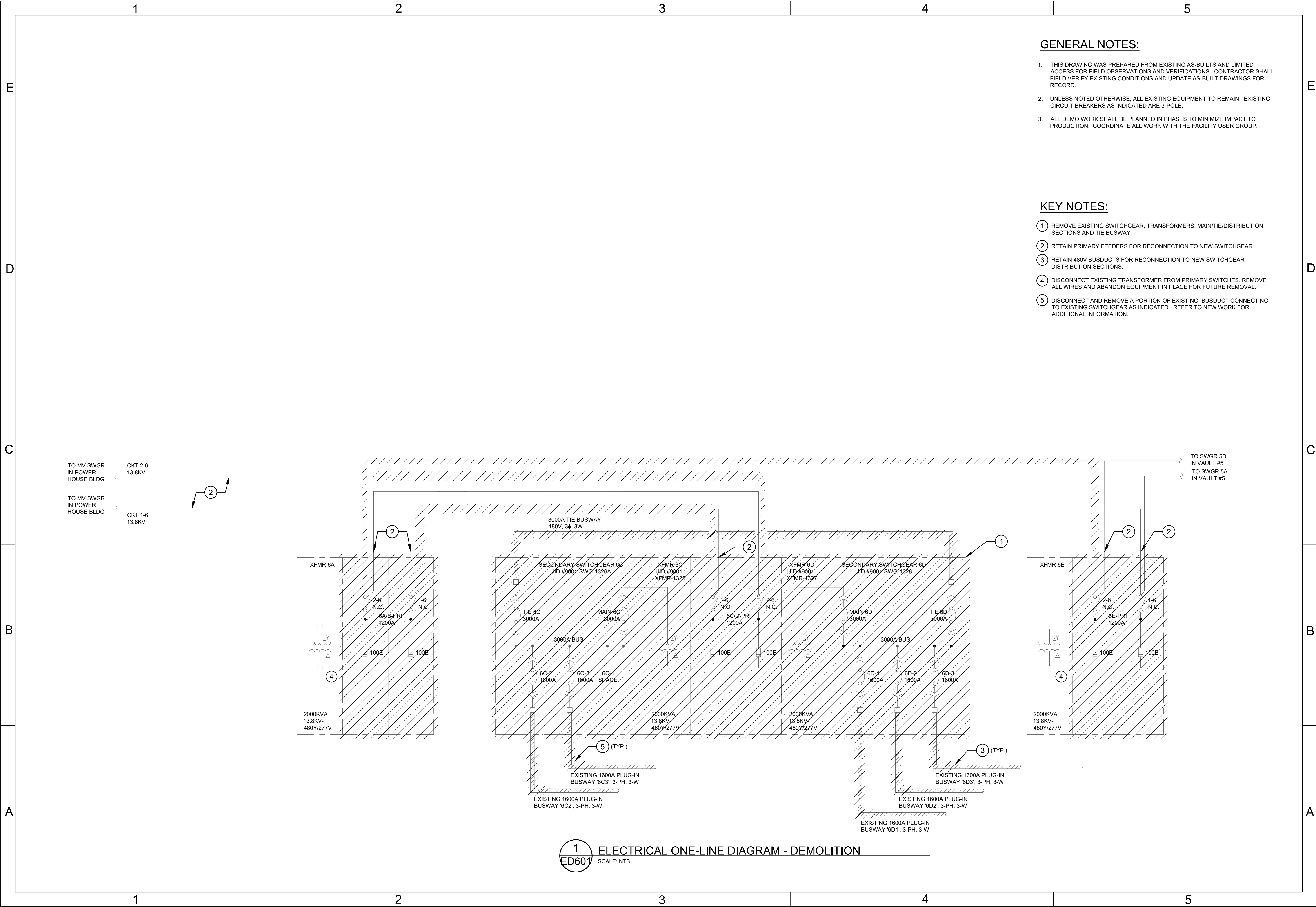
MANAGEMENT INFORMATION
ISSUE DATE: 12 DEC 2025
DESIGN GROUP: HDR
DESIGNED BY: BJ
DRAWN BY: AS
CHECKED BY: KL
DESIGN FILE LOCATION:

SHEET TITLE

ELECTRICAL
PLAN
- DEMOLITION

SHEET NUMBER

ED101



GENERAL NOTES:

- 1. THIS DRAWING WAS PREPARED FROM EXISTING AS-BUILTS AND LIMITED ACCESS FOR FIELD OBSERVATIONS AND VERIFICATIONS. CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS AND UPDATE AS-BUILT DRAWINGS FOR RECORD.
- 2. UNLESS NOTED OTHERWISE, ALL EXISTING EQUIPMENT TO REMAIN. EXISTING CIRCUIT BREAKERS AS INDICATED ARE 3-POLE.
- 3. ALL DEMO WORK SHALL BE PLANNED IN PHASES TO MINIMIZE IMPACT TO PRODUCTION. COORDINATE ALL WORK WITH THE FACILITY USER GROUP.

KEY NOTES:

- ① REMOVE EXISTING SWITCHGEAR, TRANSFORMERS, MAIN/TIE/DISTRIBUTION SECTIONS AND TIE BUSWAY.
- ② RETAIN PRIMARY FEEDERS FOR RECONNECTION TO NEW SWITCHGEAR.
- ③ RETAIN 480V BUSDUCKTS FOR RECONNECTION TO NEW SWITCHGEAR DISTRIBUTION SECTIONS.
- ④ DISCONNECT EXISTING TRANSFORMER FROM PRIMARY SWITCHES. REMOVE ALL WIRES AND ABANDON EQUIPMENT IN PLACE FOR FUTURE REMOVAL.
- ⑤ DISCONNECT AND REMOVE A PORTION OF EXISTING BUSDUCT CONNECTING TO EXISTING SWITCHGEAR AS INDICATED. REFER TO NEW WORK FOR ADDITIONAL INFORMATION.

1 ELECTRICAL ONE-LINE DIAGRAM - DEMOLITION
ED601 SCALE: NTS



AIR FORCE MATERIEL COMMAND
72 ABW CIVIL ENGINEER DIRECTORATE
TINKER AFB, OK



PROJECT INFORMATION

WYK150083
REPAIR TRANSFORMER
VAULT 6, B9001

TINKER AIR FORCE BASE, OKLAHOMA

REVISIONS	
MARK	DATE

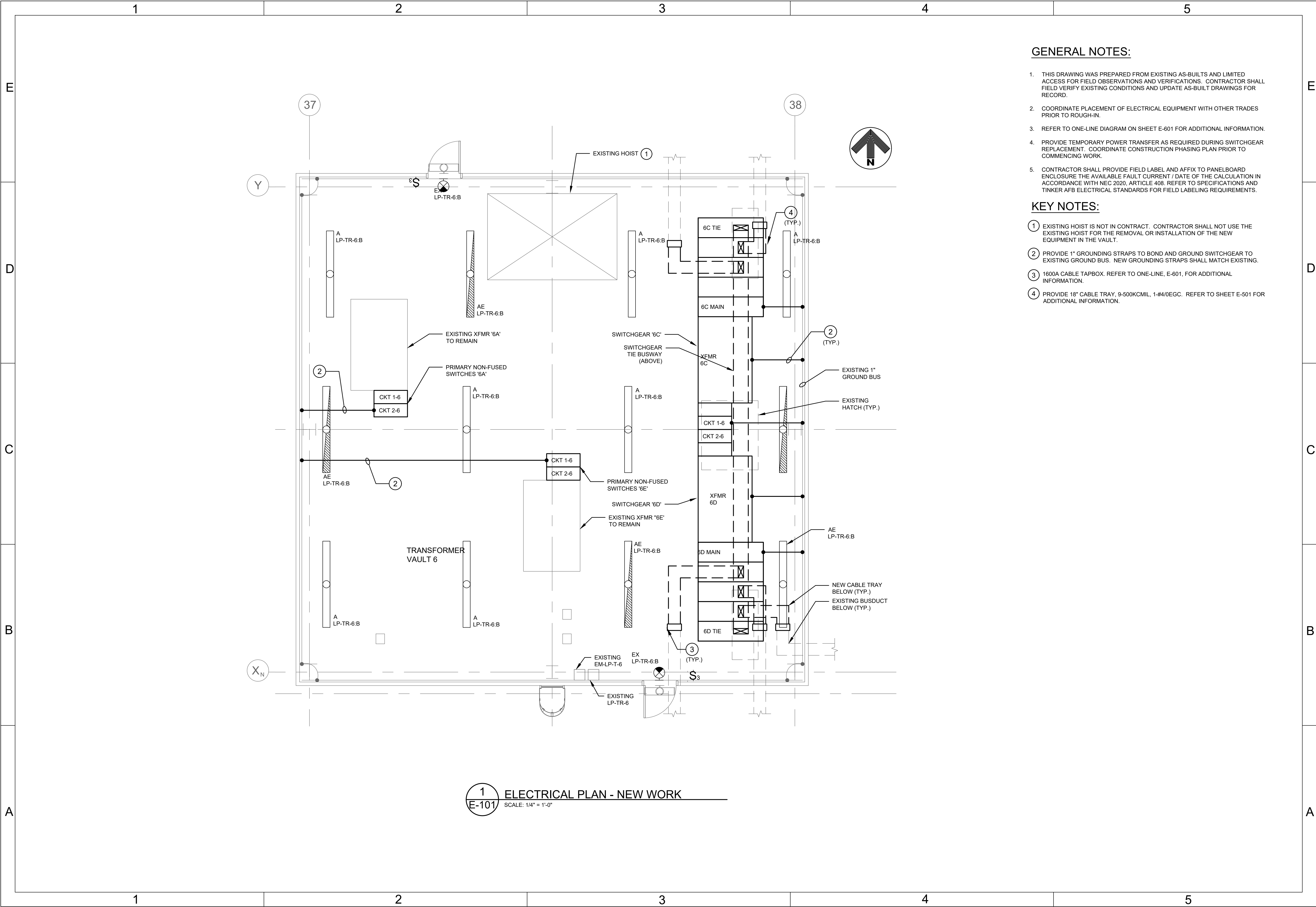
MANAGEMENT INFORMATION	
ISSUE DATE: 12 DEC 2025	DESIGN GROUP: HDR
DESIGNED BY: BJ	DRAWN BY: AS
CHECKED BY: KL	DESIGN FILE LOCATION:

SHEET TITLE

ELECTRICAL
ONE-LINE DIAGRAM
- DEMOLITION

SHEET NUMBER

ED601



GENERAL NOTES:

1. THIS DRAWING WAS PREPARED FROM EXISTING AS-BUILTS AND LIMITED ACCESS FOR FIELD OBSERVATIONS AND VERIFICATIONS. CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS AND UPDATE AS-BUILT DRAWINGS FOR RECORD.
2. COORDINATE PLACEMENT OF ELECTRICAL EQUIPMENT WITH OTHER TRADES PRIOR TO ROUGH-IN.
3. REFER TO ONE-LINE DIAGRAM ON SHEET E-601 FOR ADDITIONAL INFORMATION.
4. PROVIDE TEMPORARY POWER TRANSFER AS REQUIRED DURING SWITCHGEAR REPLACEMENT. COORDINATE CONSTRUCTION PHASING PLAN PRIOR TO COMMENCING WORK.
5. CONTRACTOR SHALL PROVIDE FIELD LABEL AND AFFIX TO PANELBOARD ENCLOSURE THE AVAILABLE FAULT CURRENT / DATE OF THE CALCULATION IN ACCORDANCE WITH NEC 2020, ARTICLE 408. REFER TO SPECIFICATIONS AND TINKER AFB ELECTRICAL STANDARDS FOR FIELD LABELING REQUIREMENTS.

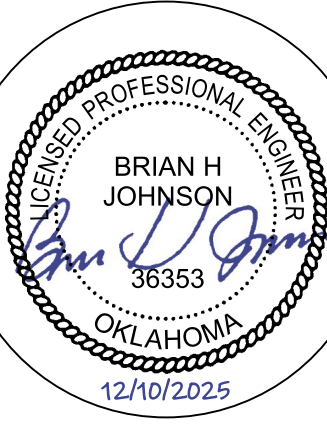
KEY NOTES:

- 1 EXISTING HOIST IS NOT IN CONTRACT. CONTRACTOR SHALL NOT USE THE EXISTING HOIST FOR THE REMOVAL OR INSTALLATION OF THE NEW EQUIPMENT IN THE VAULT.
- 2 PROVIDE 1" GROUNDING STRAPS TO BOND AND GROUND SWITCHGEAR TO EXISTING GROUND BUS. NEW GROUNDING STRAPS SHALL MATCH EXISTING.
- 3 1600A CABLE TAPBOX. REFER TO ONE-LINE, E-601, FOR ADDITIONAL INFORMATION.
- 4 PROVIDE 18" CABLE TRAY, 9-500KCMIL, 1-#4/0EGC. REFER TO SHEET E-501 FOR ADDITIONAL INFORMATION.



AIR FORCE MATERIEL COMMAND
72 ABW CIVIL ENGINEER DIRECTORATE

TINKER AFB, OK



PROJECT INFORMATION

WYK150083
REPAIR TRANSFORMER
VAULT 6, B9001

TINKER AIR FORCE BASE, OKLAHOMA

REVISIONS	MARK	DATE	DESCRIPTION

MANAGEMENT INFORMATION

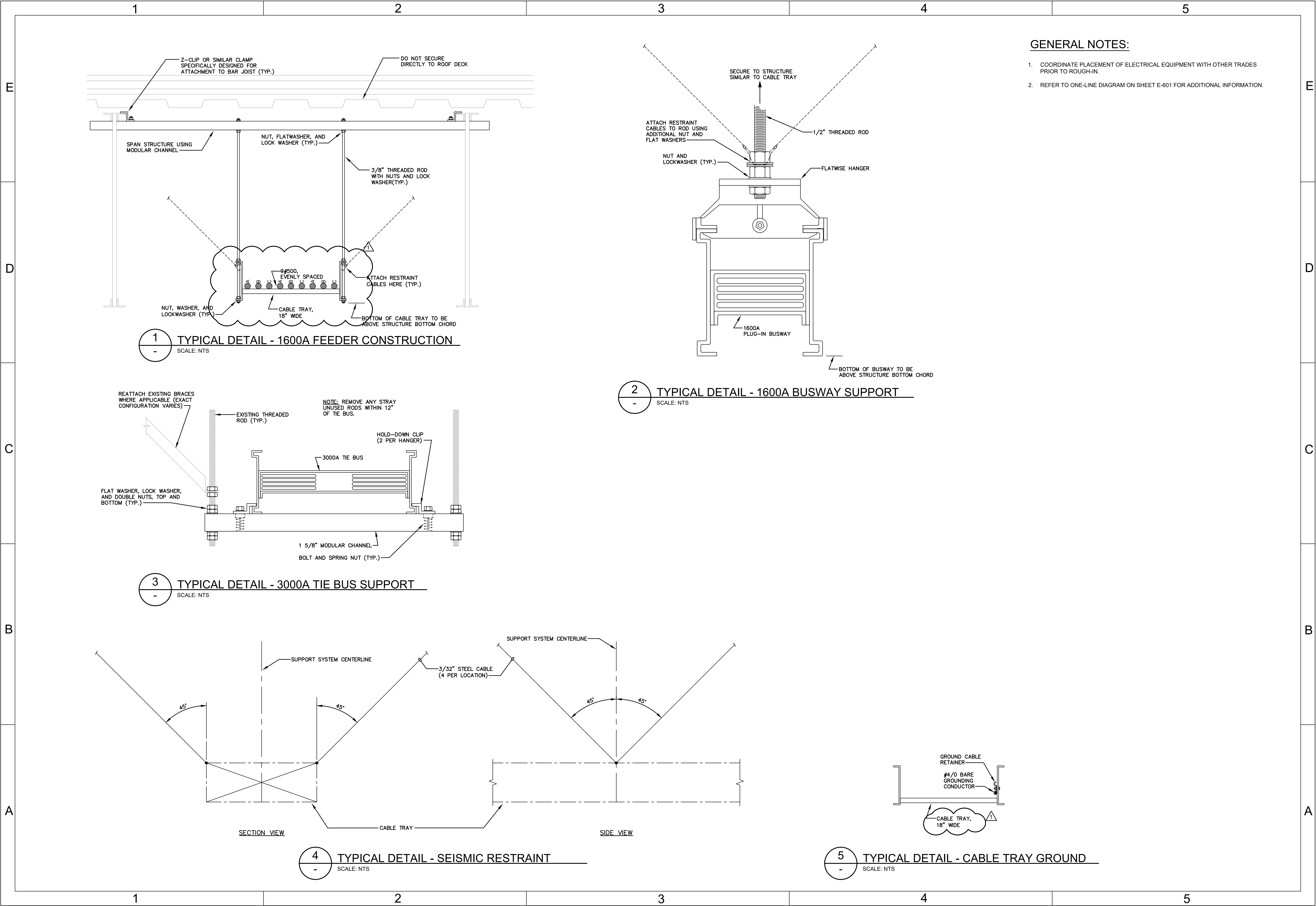
ISSUE DATE: 12 DEC 2025
DESIGN GROUP: HDR
DESIGNED BY: BJ
DRAWN BY: AS
CHECKED BY: KL
DESIGN FILE LOCATION:

SHEET TITLE

ELECTRICAL
PLAN
- NEW WORK

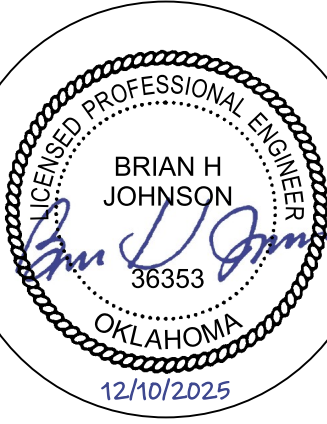
SHEET NUMBER

E-101





AIR FORCE MATERIEL COMMAND
72 ABW CIVIL ENGINEER DIRECTORATE
TINKER AFB, OK



BRIAN H. JOHNSON
36353
OKLAHOMA
12/10/2025

PROJECT INFORMATION

WWYK150083
REPAIR TRANSFORMER
VAULT 6, B9001
TINKER AIR FORCE BASE, OKLAHOMA

REVISIONS	
MARK	DATE DESCRIPTION
1	06/23/2021 REV. 1

MANAGEMENT INFORMATION

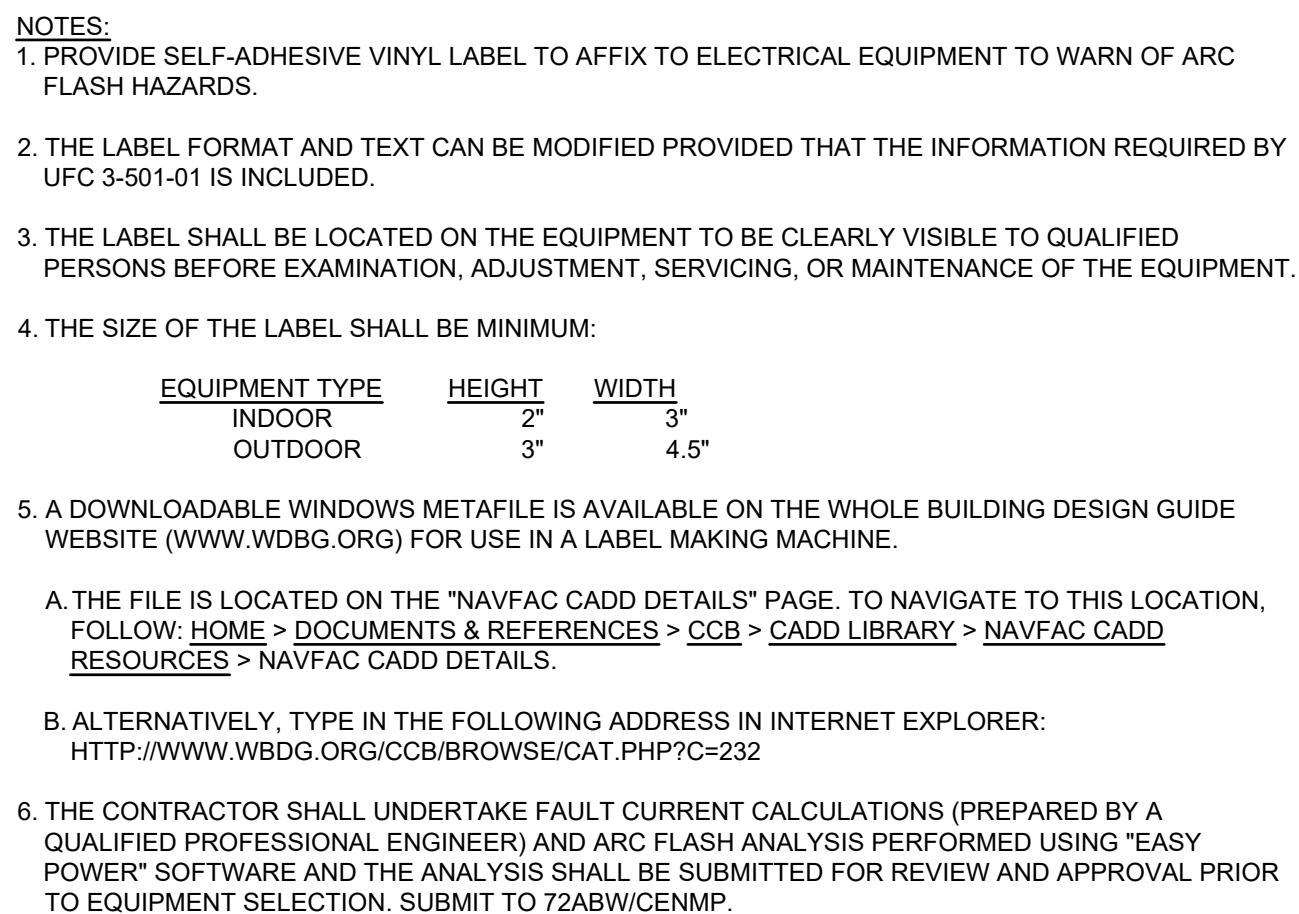
ISSUE DATE: 12 DEC 2025
DESIGN GROUP: HDR
DESIGNED BY: BJ
DRAWN BY: AS
CHECKED BY: KL
DESIGN FILE LOCATION:

SHEET TITLE

ELECTRICAL
DETAILS

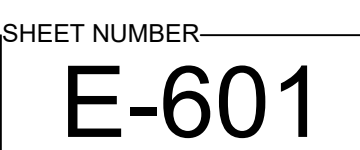
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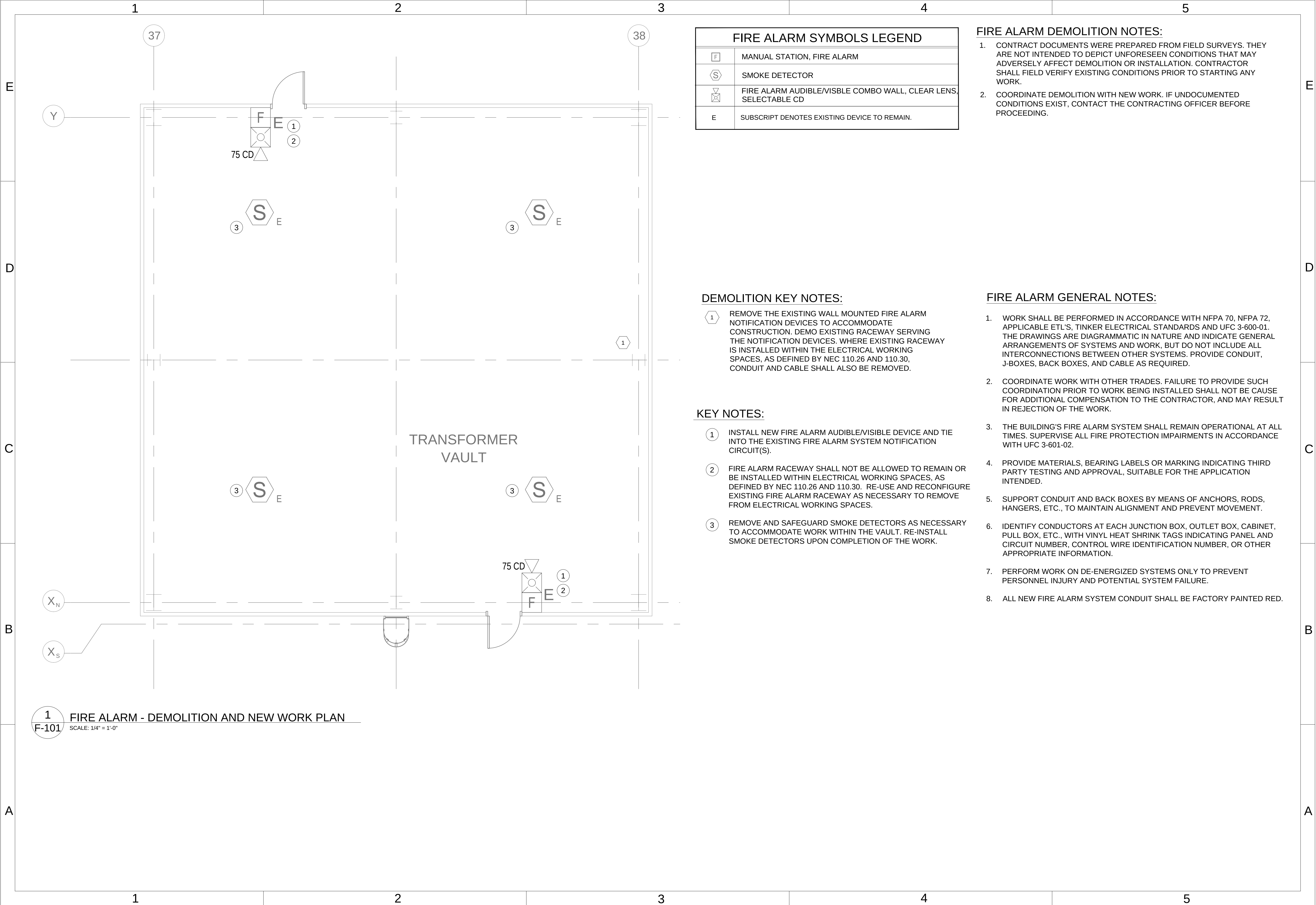
E-501



1. THIS DRAWING WAS PREPARED FROM EXISTING AS-BUILTS AND LIMITED ACCESS FOR FIELD OBSERVATIONS AND VERIFICATIONS. CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS AND UPDATE AS-BUILT DRAWINGS FOR RECORD.
2. UNLESS NOTED OTHERWISE, ALL EXISTING EQUIPMENT TO REMAIN. EXISTING CIRCUIT BREAKERS AS INDICATED ARE 3-POLE.
3. CONTRACTOR SHALL VERIFY ALL UPSTREAM OVERCURRENT PROTECTION AND SETTINGS FOR SHORT FAULT CURRENT CALCULATION AND ARC FLASH LABELING. PROVIDE APPROPRIATE ARC-FLASH LABELING PER ANALYSIS.
4. PROVIDE TEMPORARY POWER TRANSFER AS REQUIRED DURING SWITCHGEAR REPLACEMENT. COORDINATE CONSTRUCTION PHASING PLAN PRIOR TO COMMENCING WORK.
5. CONTRACTOR SHALL PROVIDE FIELD LABEL AND AFFIX TO PANELBOARD ENCLOSURE THE AVAILABLE FAULT CURRENT / DATE OF THE CALCULATION IN ACCORDANCE WITH NEC 2020, ARTICLE 408. REFER TO SPECIFICATIONS AND TINKER AFB ELECTRICAL STANDARDS FOR FIELD LABELING REQUIREMENTS.

- 1 PROVIDE SQUARE D POWER ZONE SWITCHGEAR WITH REMOTE RACKING, KIRK-KEY SWITCHING INTERLOCK, FORCED FAN COOLING, TEMPERATURE MONITORING WITH LOCAL ALARM, AND REMOTE ALARM CAPABILITIES. ALL INTERNAL WIRING TO BE DERIVED FROM INTEGRAL CPTs SO NO EXTERNAL CIRCUITS ARE REQUIRED FOR OPERATION.
- 2 CONNECT EXISTING MV FEEDERS TO NEW SWITCHGEAR PRIMARY SECTIONS.
- 3 CONNECT EXISTING 480V BUSDUCT TO CABLE TAP BOX AND EXTEND AS NEEDED. BUSDUCT EXTENSION SHALL BE SQUARE D AND COMPATIBLE WITH EXISTING.
- 4 PROVIDE T-FITTING BUSDUCT AS INDICATED FOR FUTURE EXPANSION/CONNECTIONS.
- 5 PROVIDE MEDIUM VOLTAGE RATED CABLE TO MATCH WITH EXISTING. PROVIDE PULL BOX AS REQUIRED FOR RECONNECTION TO EXISTING ONE.
- 6 PROVIDE 480V, 3PH, 3W, 1800A BUSDUCT TAPPED TO EXISTING BUSDUCT AND TERMINATE IN SWITCHGEAR. REFER TO DETAILS ON SHEET E-501 FOR ADDITIONAL INFORMATION.
- 7 IDENTIFY SWITCHGEAR GEC TEST POINT PER SECTION 3.1.4 C 1 OF THE TAFB ELECTRICAL STANDARDS IAW AFI 32-1065.





FIRE ALARM SYMBOLS LEGEND	
	MANUAL STATION, FIRE ALARM
	SMOKE DETECTOR
	FIRE ALARM AUDIBLE/VISBLE COMBO WALL, CLEAR LENS, SELECTABLE CD
E	SUBSCRIPT DENOTES EXISTING DEVICE TO REMAIN.

FIRE ALARM DEMOLITION NOTES:

1. CONTRACT DOCUMENTS WERE PREPARED FROM FIELD SURVEYS. THEY ARE NOT INTENDED TO DEPICT UNFORESEEN CONDITIONS THAT MAY ADVERSELY AFFECT DEMOLITION OR INSTALLATION. CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS PRIOR TO STARTING ANY WORK.
2. COORDINATE DEMOLITION WITH NEW WORK. IF UNDOCUMENTED CONDITIONS EXIST, CONTACT THE CONTRACTING OFFICER BEFORE PROCEEDING.

DEMOLITION KEY NOTES:

- 1 REMOVE THE EXISTING WALL MOUNTED FIRE ALARM NOTIFICATION DEVICES TO ACCOMMODATE CONSTRUCTION. DEMO EXISTING RACEWAY SERVING THE NOTIFICATION DEVICES. WHERE EXISTING RACEWAY IS INSTALLED WITHIN THE ELECTRICAL WORKING SPACES, AS DEFINED BY NEC 110.26 AND 110.30, CONDUIT AND CABLE SHALL ALSO BE REMOVED.

KEY NOTES:

- 1 INSTALL NEW FIRE ALARM AUDIBLE/VISIBLE DEVICE AND TIE INTO THE EXISTING FIRE ALARM SYSTEM NOTIFICATION CIRCUIT(S).
- 2 FIRE ALARM RACEWAY SHALL NOT BE ALLOWED TO REMAIN OR BE INSTALLED WITHIN ELECTRICAL WORKING SPACES, AS DEFINED BY NEC 110.26 AND 110.30. RE-USE AND RECONFIGURE EXISTING FIRE ALARM RACEWAY AS NECESSARY TO REMOVE FROM ELECTRICAL WORKING SPACES.
- 3 REMOVE AND SAFEGUARD SMOKE DETECTORS AS NECESSARY TO ACCOMMODATE WORK WITHIN THE VAULT. RE-INSTALL SMOKE DETECTORS UPON COMPLETION OF THE WORK.

FIRE ALARM GENERAL NOTES:

1. WORK SHALL BE PERFORMED IN ACCORDANCE WITH NFPA 70, NFPA 72, APPLICABLE ETL'S, TINKER ELECTRICAL STANDARDS AND UFC 3-600-01. THE DRAWINGS ARE DIAGRAMMATIC IN NATURE AND INDICATE GENERAL ARRANGEMENTS OF SYSTEMS AND WORK, BUT DO NOT INCLUDE ALL INTERCONNECTIONS BETWEEN OTHER SYSTEMS. PROVIDE CONDUIT, J-BOXES, BACK BOXES, AND CABLE AS REQUIRED.
2. COORDINATE WORK WITH OTHER TRADES. FAILURE TO PROVIDE SUCH COORDINATION PRIOR TO WORK BEING INSTALLED SHALL NOT BE CAUSE FOR ADDITIONAL COMPENSATION TO THE CONTRACTOR, AND MAY RESULT IN REJECTION OF THE WORK.
3. THE BUILDING'S FIRE ALARM SYSTEM SHALL REMAIN OPERATIONAL AT ALL TIMES. SUPERVISE ALL FIRE PROTECTION IMPAIRMENTS IN ACCORDANCE WITH UFC 3-601-02.
4. PROVIDE MATERIALS, BEARING LABELS OR MARKING INDICATING THIRD PARTY TESTING AND APPROVAL, SUITABLE FOR THE APPLICATION INTENDED.
5. SUPPORT CONDUIT AND BACK BOXES BY MEANS OF ANCHORS, RODS, HANGERS, ETC., TO MAINTAIN ALIGNMENT AND PREVENT MOVEMENT.
6. IDENTIFY CONDUCTORS AT EACH JUNCTION BOX, OUTLET BOX, CABINET, PULL BOX, ETC., WITH VINYL HEAT SHRINK TAGS INDICATING PANEL AND CIRCUIT NUMBER, CONTROL WIRE IDENTIFICATION NUMBER, OR OTHER APPROPRIATE INFORMATION.
7. PERFORM WORK ON DE-ENERGIZED SYSTEMS ONLY TO PREVENT PERSONNEL INJURY AND POTENTIAL SYSTEM FAILURE.
8. ALL NEW FIRE ALARM SYSTEM CONDUIT SHALL BE FACTORY PAINTED RED.

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ISSUE DATE:	12 DEC 2025	MARK	DESCRIPTION
DESIGN GROUP:	HDR <td></td> <td></td>		
DESIGNED BY:	DE <td></td> <td></td>		
DRAWN BY:	JPM <td></td> <td></td>		
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DESIGN FILE LOCATION:			

SHEET TITLE

FIRE ALARM -
DEMOLITION AND
NEW WORK PLAN

SHEET NUMBER

F-101